



ARRAY

CCE 103/L

○ Objectives

- Understand the fundamental concepts of arrays in Java
- Apply different techniques for working with arrays effectively
- Analyze and solve common programming problems using arrays



● Java Arrays

Java provides a data structure, the array, which stores a fixed-size sequential collection of elements of the same type. An array is used to store a collection of data, but it is often more useful to think of an array as a collection of variables of the same type.

It is a group of like-typed variables that are referred to by a common name. Arrays of any type can be created and may have one or more dimensions. A specific element in an array is accessed by its index. Arrays offer a convenient means of grouping related information.



○ Declaring Array Variables

To use an array in a program, you must declare a variable to reference the array, and you must specify the type of array the variable can reference.

dataType[] arrayRefVar; // preferred way.

or

dataType arrayRefVar[]; // works but not preferred way.



○ Declaring Array Variables

Example:

double[] myList; // preferred way.

or

double myList[]; // works but not preferred way.



One-Dimensional Array

To create an array, you first must create an array variable of the desired type. The general form of a one dimensional array declaration is

type var-name[];

Here, type declares the base type of the array. The *base type* determines the data type of each element that comprises the array. Thus, the base type for the array determines what type of data the array will hold.

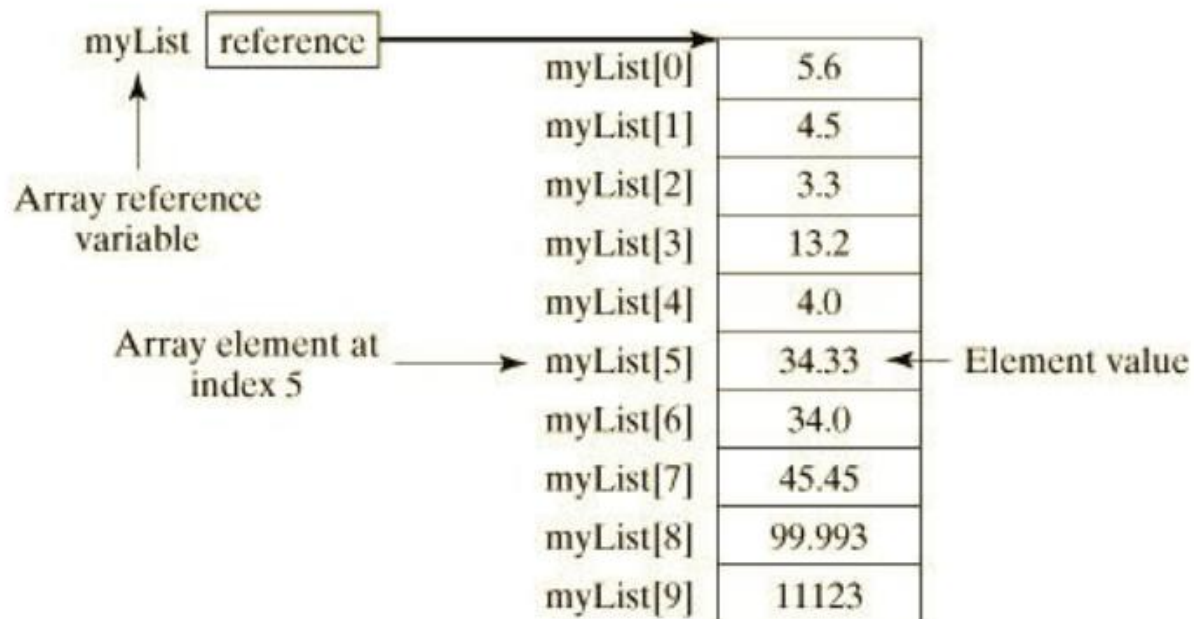
int month_days[];

array-var = new type[size];

month_days = new int[12];



One-Dimensional Array



One-Dimensional Array

```
int month_days[];  
month_days = new  
int[12];  
month_days[0] = 31;  
month_days[1] = 28;  
month_days[2] = 31;  
month_days[3] = 30;  
month_days[4] = 31;  
month_days[5] = 30;
```

```
month_days[6] = 31;  
month_days[7] = 31;  
month_days[8] = 30;  
month_days[9] = 31;  
month_days[10] = 30;  
month_days[11] = 31;  
System.out.println("April  
has " + month_days[3] +  
" days.");
```



○ One-Dimensional Array

Example:

// An improved version of the previous program.

```
class AutoArray {  
    public static void main(String args[]) {  
        int month_days[] = { 31, 28, 31, 30, 31, 30, 31, 31, 30, 31,  
30, 31 };  
        System.out.println("April has " + month_days[3] + " days.");  
    }  
}
```



One-Dimensional Array

Example:

// Average an array of values.

```
class Average {  
    public static void main(String args[]) {  
        double nums[] = {10.1, 11.2, 12.3, 13.4, 14.5};  
        double result = 0;  
        int i;  
        for(i=0; i<5; i++){  
            result = result + nums[i];  
            System.out.println("Average is " + result / 5);}  
        }  
    }
```



One-Dimensional Array

Example:

```
double[] myList = {1.9, 2.9, 3.4, 3.5};
```

```
// Print all the array elements
```

```
for(int i = 0; i < myList.length; i++){  
    System.out.println(myList[i] + " "); }  
// Summing all elements
```

```
double total = 0;
```

```
for(int i = 0; i < myList.length; i++){  
    total += myList[i]; }  
System.out.println("Total is " + total);
```

```
// Finding the largest element
```

```
double max = myList[0];
```

```
for(int i = 1; i < myList.length; i++){
```

```
    if(myList[i] > max)
```

```
        max = myList[i];
```

```
}
```

```
System.out.println("Max is " +  
max);
```



○ **One-Dimensional Array**

What is the output of this program?

```
int[] no={1,2,3,4,5};
```

```
for(int i=0;i<5;i++)
```

```
    no[i]+=10;
```

```
for(int i=0;i<5;i++)
```

```
System.out.print(no[i] + " ");
```



○ One-Dimensional Array

What is the output of this program?

```
int numbers[] = {1, 2, 3, 4, 5, 6, 7, 8,9,10};  
int sum=0;  
for (int i = 0; i <numbers.length ; i++)  
    sum += numbers[i];  
System.out.println("The sum of these numbers is " +  
sum);
```





One-Dimensional Array

Example:

```
import java.util.Scanner;
```

```
class SumDemo{
```

```
    public static void main(String args[]){
```

```
        Scanner scanner = new Scanner(System.in);
```

```
        int[] array = new int[10];
```

```
        int sum = 0;
```

```
        System.out.println("Enter the elements:");
```

```
        for (int i=0; i<10; i++)    {
```

```
            array[i] = scanner.nextInt();
```

```
        }
```

```
        for( int num : array) {
```

```
            sum = sum+num;
```

```
        }
```

```
        System.out.println("Sum of  
array elements is:"+sum);
```

```
    }
```

```
}
```





Exercise

Write a Java program using array that will determine whether the numbers from 1-10 is odd or even.

Output:

1 is an odd number.
2 is an even number.
3 is an odd number.
4 is an even number.
5 is an odd number.
6 is an even number.
7 is an odd number.
8 is an even number.
9 is an odd number.
10 is an even number.



○ **public class FindEvenOrOddNumber {**
public static void main(String[] args) {
 int[] numbers = new int[]{1,2,3,4,5,6,7,8,9,10};
 for(int i=0; i < numbers.length; i++){
 if(numbers[i]%2 == 0){
 System.out.println(numbers[i] + " is even number.");
 else {
 System.out.println(numbers[i] + " is odd number.");
 }
 }}





Assignment

- Trace the program you have performed earlier. Use a clean short bond paper upon tracing.

